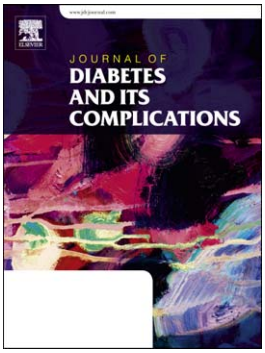

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Clinical and metabolic response to flaxseed oil omega-3 fatty acids supplementation in patients with diabetic foot ulcer: a randomized, double-blind, placebo-controlled trial

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Running Title: Omea-3 fatty acids and diabetic foot

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ABSTRACT

Background: Data on the effects of flaxseed oil omega-3 fatty acids supplementation on wound healing and metabolic status in subjects with diabetic foot ulcer (DFU) are scarce.

Objective: This study was conducted to evaluate the effects of flaxseed oil omega-3 fatty acids supplementation on wound healing and metabolic status in subjects with DFU.

Methods: The current randomized, double-blind, placebo-controlled trial was conducted among 60 subjects (aged 40-85 years old) with grade 3 DFU. Subjects were randomly allocated into two groups (30 subjects each group) to receive either 1000 mg omega-3 fatty acids from flaxseed oil supplements or placebo twice a day for 12 weeks.

Results: After the 12-week intervention, compared with the placebo, omega-3 fatty acids supplementation resulted in significant decreases in ulcer length (-2.0±2.3 vs. -1.0±1.1 cm, P=0.03), width (-1.8±1.7 vs. -1.0±1.0 cm, P=0.02) and depth (-0.8±0.6 vs. -0.5±0.5 cm, P=0.01). Additionally, significant reductions in serum insulin concentrations (-4.4±5.5 vs. +1.4±8.3 µIU/mL, P=0.002), homeostasis model of assessment-estimated insulin resistance (-2.1±3.0 vs. +1.0±5.0, P=0.005) and HbA1c (-0.9±1.5 vs. -0.1±0.4%, P=0.01), and a significant rise in the quantitative insulin sensitivity check index (+0.01±0.01 vs. -0.005±0.02, P=0.002) were seen following supplementation with omega-3 fatty acids compared with the placebo. In addition, omega-3 fatty acids supplementation significantly decreased serum high sensitivity C-reactive protein (hs-CRP) (-25.5±31.5 vs. -8.2±18.9 µg/mL, P=0.01), and significantly increased plasma total antioxidant capacity (TAC) (+83.5±111.7 vs. -73.4±195.5 mmol/L, P<0.001) and glutathione (GSH) concentrations (+60.7±140.2 vs. -15.5±129.7 µmol/L, P=0.03) compared with the placebo.

Conclusions: Overall, omega-3 fatty acids supplementation for 12 weeks among subjects with DFU had beneficial effects on parameters of ulcer size, markers of insulin metabolism, serum hs-CRP, plasma TAC and GSH levels. In addition, flaxseed oil omega-3 fatty acids

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may have played an indirect role in wound healing due to its effects on improved metabolic profiles.

Keywords: Omega-3 fatty acids administration, wound healing, metabolic status, diabetic foot

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1. Introduction

Diabetes-related foot complications have been reported as the single most common cause of morbidity among diabetic subjects (Lim *et al.*, 2017). Although contemporary data on diabetic foot ulcer (DFU) prevalence are scarce, most studies have reported the its annual incidence in the diabetic population around 2% (Jarbrink *et al.*, 2016; Stoekenbroek *et al.*, 2017).Up to 66% of the subjects admitted to the hospital with DFU have diabetic foot osteomyelitis (DFO) (Hartemann-Heurtier & Senneville, 2008). Increased inflammatory markers are closely related to the pathogenesis and complications of diabetes (Navarro-Gonzalez & Mora-Fernandez, 2008), which in turn may increase the incidence of DFU. In addition, the balance between pro- and anti-inflammatory processes is vital in the development of wound healing (Weigelt *et al.*, 2009). Several studies have also suggested that hyperglycemia, insulin resistance, dyslipidemia and oxidative stress play a main function in the pathogenesis of DFU (Karadurmus *et al.*, 2010; Sytze Van Dam *et al.*, 2013).

Wound healing is a complex and biological process that occurs in at least three overlapping stages, the first being inflammation (Werner & Grose, 2003; Falanga, 2005). Molecular and cellular processes during the inflammatory stage of skin healing are initiated and amplified to a large degree by a group of protein mediators known as pro-inflammatory cytokines (Efron & Moldawer, 2004). Pro-inflammatory cytokines synthesis and activity are affected by the concentration of omega-3 fatty acids. In a study by McDaniel et al.(McDaniel *et al.*, 2008) it was observed that omega-3 fatty acids supplementation in a healthy human population for 4 weeks resulted in increased production of pro-inflammatory cytokines, including interleukin-1β (IL-1β) at wound sites and thus, depending on the clinical context, have noninvasive, therapeutic potential to affect cutaneous wound healing. In addition, omega-3 fatty acids supplementation in microenvironment of acutehuman wounds improved wound healing (McDaniel *et al.*, 2011). Almost all earlier studies have evaluated the effects of omega-3 fatty

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acids from fish oil supplements on wound healing among patients without DFU. Furthermore, data on the effects of omega-3 fatty acids on wound healing conflicting. About the effects of omega-3 fatty acids on inflammatory factors, a meta-analysis study has demonstrated that marine-derived omega-3 supplementation had a significant lowering effect on C-reactive protein (CRP), IL-6, and tumor necrosis factor α (TNF-α) levels (Li *et al.*, 2014). We have previously shown that taking omega-3 fatty acids from flaxseed oil (1 g/day) for 12 weeks among diabetic nephropathy (DN) patients had favorable effects on markers of insulin metabolism, serum triglycerides and VLDL-cholesterol, but did not affect biomarkers of inflammation and oxidative stress (Soleimani *et al.*, 2015). However, no significant change in insulin levels was seen following supplementation with 3.6 g of omega-3 fatty acid among patients with chronic kidney disease (CKD) for 10 weeks (Guebre-Egziabher *et al.*, 2013).

The beneficial effects of omega-3 fatty acids on wound healing may be related to its anti-inflammatory and antioxidant action (Zhuang *et al.*, 2013). Moreover, modulating the secretion of adipocytokines and inflammatory markers and enhancing fatty acid β-oxidation by omega-3 fatty acids (Liu *et al.*, 2013) may improve metabolic profiles in patients with DFU. As there is evidence that taking omega-3 fatty acids may accelerate wound healing and has anti-diabetic effect, we hypothesized that flaxseed oil omega-3 fatty acids supplementation might help subjects with DFU to heal their wound faster, and have a better metabolic profiles, and biomarkers of inflammation and oxidative stress. The aim of the current study, therefore, was to determine the effects of flaxseed oil omega-3 fatty acids supplementation on wound healing and metabolic profiles in subjects with DFU.

2. Methods

2.1. Trial design and participants

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The current study was a randomized double-blind placebo-controlled clinical trial, registered in the Iranian registry of clinical trials (http://www.irct.ir: IRCT201604025623N72), performed at an infectious clinic affiliated to Kashan University of Medical Sciences (KUMS), Kashan, Iran, between April 2016 and July 2016. This study was conducted according to the principals of the Declaration of Helsinki and the study protocol was approved by the ethics committee of KUMS. All participants were informed about the aims and protocol of the study. Written informed consent was obtained from all subjects prior to the intervention. In the current study, 60 subjects with grade 3 DFU according to "Wagner-Meggitt's" criteria aged 40-85 years who had deep ulcer with cellulitis (Jain, 2012) and normal distal pulse were included. Exclusion criteria were as follows: grade 3 DFU with abscess formation and osteomyelitis, subjects with significant arterial disease, major deformities e.g. Charcot's neuroarthropathy, pregnant and breastfed patients, taking antioxidant and/or anti-inflammatory supplements such as vitamin E, vitamin C and omega-3 fatty acids and change in consuming medications throughout the study. The diagnosis of abscess and osteomyelitis was conducted by routine imaging (sonography and MRI, respectively).

2.2. Study design

Firstly, all persons were matched according to gender (males: 23 and 7 females in each group), type and dosage of medications, duration of diabetes mellitus (DM), pre-intervention BMI (<25 and ≥25 kg/m^2) and age (<55 and ≥55 years). Then, participants were randomized to the two treatment arms to receive either 1000 mg/day omega-3 fatty acid from flaxseed oil or placebo (n=30 in each group) twice a day for 12 weeks. Omega-3 fatty acids supplements and placebos (paraffin) were made by Barij Pharmaceutical Company (Kashan, Iran). In addition, all persons underwent a similar treatment protocol based on the international

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consensus on the diabetic foot (Schaper *et al.*, 2003; Bader, 2008), which included ciprofloxacin (Cipro) 400 mg IV twice a day plus clindamycin 900 mg IV three times per day for 3 weeks. Individuals were requested not to change their usual physical activity and not to take any nutritional supplements during the 12-week intervention. All subjects were completed 3-day food records and 3-physical activity records at baseline, weeks 3, 6 and 9 of the intervention and end-of-trial. Daily macro- and micro-nutrient intakes were analyzed by nutritionist IV software (First Databank, San Bruno, CA). In the current study, physical activity was described as metabolic equivalents (METs) in hours per day. To determine the METs for each patient, we multiplied the times (in hour per day) reported for each physical activity by its related METs coefficient by standard tables (Ainsworth *et al.*, 2000).

2.3. Treatment adherence

Every month, persons were given enough supplements to last until 3 days after their next scheduled visit and were instructed to return all the unused supplements at each visit. To determine the compliance of remaining supplements were counted and subtracted from the amount of supplements provided to the participants.

2.4. Assessment of anthropometric measures

Weight and height of participants were determined in a fasting status using a standard scale (Seca, Hamburg, Germany) at baseline and after the 12-week treatment. BMI was calculated as weight in kg divided by height in meters squared.

2.5. Assessment of outcomes

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Wound healing and glucose homeostasis parameters were considered as the primary outcomes measurements, and lipid profiles, biomarkers of inflammation and oxidative stress were considered as the secondary outcomes end-points.

2.6. Clinical assessment

Mean ulcer area was estimated as the product of longest measured length times longest perpendicular to length. Ulcer volume was estimated as the estimated area times the deepest ulcer depth (Santamaria *et al.*, 2012). Depth ulcer was defined as deep and was measured using a sterile blunt nasal probe (Santamaria *et al.*, 2012). Infection was diagnosed if edema, erythema, discharge, regional lymph node enlargement, pain, or fever was present (Doupis & Veves, 2008). Participants were considered positive for infection if at least two of the above listed features were present. Each foot was evaluated and graded according to the Wagner classification. Frequency of wound measurement was conducted at baseline, weeks 1, 3, 4 and 8 of the intervention and end-of-trial**.**

2.7. Biochemical assessment

At baseline and after the 12-week intervention, 10 mL blood samples were obtained from each patient at Kashan reference laboratory in a fating status. Then, the samples were stored at -80$^{\circ C}$ before analysis. Serum insulin concentrations were quantified by available ELISA kit (DiaMetra, Milano, Italy) with intra- and inter-assay coefficient variances (CVs) of 2.7 and 4.6%, respectively. The homeostasis model of assessment-insulin resistance (HOMA-IR) and the quantitative insulin sensitivity check index (QUICKI) were calculated using the suggested formulas (Pisprasert *et al.*, 2013). Hemoglobin A1c ($HbA1_C$) levels in the whole blood were determined by Glycomat kit (BiocodeHycel, Massy, France) using the method of exchange chromatography. Enzymatic kits of Pars Azmun (Tehran, Iran) were used to assess fasting

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plasma glucose (FPG), serum triglycerides, VLDL-, total-, LDL- and HDL-cholesterol concentrations with inter- and intra-assay CVs less than 5%. Serum high sensitivity C-reactive protein (hs-CRP) concentrations were evaluated by a commercial ELISA kit (LDN, Nordhorn, Germany) with intra- and inter-assay CVs of 3.6 and 5.9%, respectively. The plasma nitric oxide (NO) values were assessed using Griess method (Tatsch *et al.*, 2011). Plasma total antioxidant capacity (TAC) concentrations by the use of ferric reducing antioxidant power developed by Benzie and Strain (Benzie & Strain, 1996), total glutathione (GSH) by the method of Beutler et al. (Beutler & Gelbart, 1985) and malondialdehyde (MDA) concentrations by the thiobarbituric acid reactive substances spectrophotometric test (Janero, 1990) were evaluated with inter- and intra-assay CVs less than 5%.

2.8. Sample size

To estimate the sample size, we used a randomized clinical trial sample size formula (Choi, 1998) where type one (α) and type two errors (β) were 0.05 and 0.20 (power=80%), respectively. Based on a previous study (Soleimani *et al.*, 2015), we used a standard deviation (SD) of 0.6 and a difference in mean (d) of 0.5, considering HOMA-IR as the key variable. The calculation indicated 25 participants were needed in each group. Assuming a dropout of 5 participants per group, the final sample size was determined to be 30 participants per group.

2.9. Randomization

Randomization assignment was conducted using computer-generated random numbers. Randomization and allocation were concealed from the researchers and participants until the final analyses were completed. The randomized allocation sequence, enrolling participants and allocating them to interventions were carried out by a trained staff at the clinic.

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2.10. Statistical analysis

To ensure the normal distribution of variables, the Kolmogorov-Smirnov test was used. The analyses were carried out based on intention-to-treat (ITT) principle. To detect differences in anthropometric measures as well as in dietary intakes between the two groups, we used independent samples Student's t test. To determine the effects of omega-3 fatty acids supplementation on ulcer size, glucose homeostasis parameters, lipid profiles, biomarkers of inflammation and oxidative stress, we used one-way repeated measures analysis of variance. Differences in proportions were evaluated by Chi square test. To control confounding variables including baseline values, age and baseline BMI, we used analysis of covariance (ANCOVA). P-values <0.05 were considered statistically significant. All statistical analyses were done by the use of the Statistical Package for Social Science version 18 (SPSS Inc., Chicago, Illinois, USA).

3. Results

Among subjects in the placebo group, 2 persons and in the omega-3 fatty acids group, 3 persons (withdrawn due to personal reasons) was excluded (Fig.1). However, as the analysis was conducted based on the ITT principle, all 60 persons (30 persons in each group) were included in the final analyses. No side effects were reported following the consumption of flaxseed oil omega-3 fatty acids supplements in patients with DFU throughout the study.

Gender distribution, mean age, height, weight and BMI at baseline and after the 12-week intervention, METs at baseline and end-of-trial, and insulin and metformin therapy were not statistically different between the two groups (Table 1).

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Based on the 3-day dietary records obtained at baseline, end-of-trial and throughout the study, we found no significant difference in mean dietary macro- and micro-nutrient intakes between the two groups (Table 2).

After the 12-week intervention, compared with the placebo, omega-3 fatty acids supplementation led to significant decreases in ulcer length (-2.0±2.3 vs. -1.0±1.1 cm, P=0.03), width (-1.8±1.7 vs. -1.0±1.0 cm, P=0.02) and depth (-0.8±0.6 vs. -0.5±0.5 cm, P=0.01) (Table 3). Additionally, significant reductions in serum insulin concentration (-4.4±5.5 vs. +1.4±8.3 µIU/mL, P=0.002) and HOMA-IR (-2.1±3.0 vs. +1.0±5.0, P=0.005), and a significant rise in QUICKI (+0.01±0.01 vs. -0.005±0.02, P=0.002) were seen following supplementation with omega-3 fatty acids compared with the placebo. In addition, omega-3 fatty acids supplementation significantly decreased serum hs-CRP (-25.5±31.5 vs. -8.2±18.9 µg/mL, P=0.01), and significantly increased plasma TAC (+83.5±111.7 vs. -73.4±195.5 mmol/L, P<0.001) and GSH concentrations (+60.7±140.2 vs. -15.5±129.7 µmol/L, P=0.03) compared with the placebo. There were no significant changes in lipid profiles and other biomarkers of inflammation and oxidative stress between the two groups.

There was a significant difference in baseline values of plasma NO (P=0.002) and MDA (P<0.001) between the two groups. Therefore, we adjusted the analysis for baseline values of biochemical variables, age and baseline BMI. When we adjusted the analysis for baseline values of biochemical variables, age and baseline BMI, no significant change in our findings (Table 4).

4. Discussion

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In this study, which to our knowledge is the first of its kind, we assessed effects of omega-3 fatty acids supplementation from flaxseed oil on wound healing, glycemic control, lipid profiles, and biomarkers of inflammation and oxidative stress among subjects with DFU. We found that omega-3 fatty acids supplementation for 12 weeks among persons with DFU had beneficial effects on parameters of ulcer size, insulin metabolism, hs-CRP, plasma TAC and GSH levels, but did not had any effect on lipid profiles and other biomarkers of inflammation and oxidative stress. It must be kept in mind that one might conclude that the presence of antibiotic resistant organisms might impair response to wound healing. As at baseline, all patients were matched based on gender, type and dosage of antibiotics, duration of diabetes mellitus, pre-intervention BMI and age, it is expected that the effects of antibiotic resistant organisms are same in both groups. Nonetheless, we believe that further studies are needed to confirm our findings. Ulcer location may have an indirect role on wound healing and metabolic profiles. However, findings of our study have shown that ulcer location were not statistically different between the two groups.

We found that omega-3 fatty acids supplementation in subjects with DFU for 12 weeks resulted in a significant improvement in wound healing compared with the placebo. However, data on the effects of omega-3 fatty acids from flaxseed oil on wound healing in human studies are scarce. McDaniel et al.(McDaniel *et al.*, 2008) demonstrated that omega-3 fatty acids supplementation in a healthy human population for 4 weeks resulted in increased production of pro-inflammatory cytokines, including IL-1β at wound sites and thus, depending on the clinical context, have noninvasive, therapeutic potential to affect cutaneous wound healing. IL-1 assists in regulating fibroblast proliferation and collagen synthesis (Efron & Moldawer, 2004) therefore, it can be posited that its initial upregulation at the wound site, as a result of omega-3 fatty acids supplementation, could be a pathway to

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regulate collagen formation. Improved production of healthy collagen is advantageous for efficient healing of skin with minimal scarring and for providing strength for connective tissues such as ligaments (Jia & Turek, 2005). Moreover, initial IL-1 expression increases keratinocyte growth, which is important for reepithelialization (Sauder *et al.*, 1990), induces the production of IL-6 and has a stimulatory effect on angiogeneses. Likewise, omega-3 fatty acids intake in microenvironment of acutehuman wounds improved wound healing (McDaniel *et al.*, 2011). In addition, a previous study that measured wound healing in dog models found a longer time for reepithelialization in surgical wounds after omega-3 fatty acids supplementation when compared with other dietary alterations (Mooney *et al.*, 1998). Previous wound studies have shown omega-3 supplementation to increase wound inflammation. In addition, wounds in people with diabetes already show increased and persistent inflammation. However, in the current study, systemic inflammation was decreased with omega-3 supplementation, but local inflammation was not measured. Therefore, further studies are needed to evaluate omega-3 supplementation on local inflammation in patients with DFU. Few studies have reported beneficial effects of topical omega-3 fatty acids on wound healing. For instance, in a study by Cardoso et al. (Cardoso *et al.*, 2004) it was reported that topical administration of omega-3 fatty acids to surgical wounds of mice significantly delayed wound closure in the first 10 days after surgery. Furthermore, in the above mentioned study, topical omega-3 fatty acids induced a peak in nitric oxide at 3 hours postsurgery and an intense deposition of extracellular matrix after 5 days of treatment, which in turn suggest a relevant role and potential therapeutic implication for fatty acids on skin wound healing (Cardoso *et al.*, 2004). Unlike, Ruthig et al.(Ruthig & Meckling-Gill, 1999) demonstrated an improved reconstitution of epithelial integrity with both omega-3 and omega-6 fatty acids-treated intestinal cells in rats following mucosal injury. The findings of previous studies concerning the effects of omega-3 fatty acids on pro-inflammatory cytokines

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levels may be related to their potential ability to affect epidermal wound healing due to their influence on pro-inflammatory factors and thus, cell types that are important to the healing process (McDaniel *et al.*, 2008).

We found that omega-3 fatty acids supplementation, compared with the placebo, in subjects with DFU for 12 weeks had potential to decrease serum insulin, HOMA-IR and HbA1c levels, but did not affect FPG and lipid profiles. Interventional studies about omega-3 fatty acids supplementation on insulin metabolism and lipid concentrations gave controversial results that some showing a positive and some a negative effect. Our previous study among patients with DN have shown that omega-3 fatty acids supplementation from flaxseed oil (1 g/day) for 12 weeks improved insulin metabolism, serum triglycerides and VLDL-cholesterol, but did not affect biomarkers of inflammation and oxidative stress (Soleimani *et al.*, 2015). In addition, omega-3 fatty acids supplementation in obese subjects for 12 weeks resulted in decreased concentrations of insulin, triglycerides and several inflammatory markers (Polus *et al.*, 2016). Hutchins et al. (Hutchins *et al.*, 2013) shown that daily flaxseed oil consumption among overweight or obese subjects with pre-diabetes for 12 weeks decreased insulin concentrations and improved insulin sensitivity. Significant decreases in glycemia, insulin, insulin resistance and triglycerides, and a significant increase in HDL-cholesterol levels were also observed following supplementation with 1 g of omega-3 fatty acids three times a day for 18 months in overweight/obese patients with impaired fasting glucose; however, it did not affect other lipid profiles (Derosa *et al.*, 2016). Furthermore, our findings were in agreement with what reported by Albert et al. (Albert *et al.*, 2014) that reported that higher omega-3 fatty acids index was associated with increased insulin sensitivity and a more favorable metabolic profile in middle-aged overweight men. However, some researchers did not observe any beneficial effects of omega-3 fatty acids

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supplementation on glycemic control. For instance, dietary EPA+DHA supplementation for 6 months did not improve peripheral glucose disposal, insulin secretion, or skeletal muscle mitochondrial function in insulin-resistant nondiabetic subjects (Lalia *et al.*, 2015). In addition, flaxseed oil intake in patients with T2DM had no effect on fasting serum glucose, insulin or HbA1c levels (Barre *et al.*, 2008). The different findings might be related to differences in sample size (Rhee & Brunt, 2011), methods of delivering flaxseed supplementation (eg, baked in bread, flaxseed-derived lignans or capsules) (Bloedon *et al.*, 2008; Rhee & Brunt, 2011; Soleimani *et al.*, 2017), or participant disease states (eg, T2DM, hypercholesterolemia) (Pan *et al.*, 2007; Bloedon *et al.*, 2008). Furthermore, different results might be related to the dosage and the formulations used or possibly some other component associated with flaxseed oil as well as duration of the study. In the current study, beneficial effects of taking omega-3 fatty acids on insulin metabolism may be mediated by reduction of free fatty acids (Albert *et al.*, 2014), modulating function of peroxisome proliferator-activated receptors and sterol regulatory binding proteins (Kliewer *et al.*, 1997). Another mechanism was suggested by Oh DY et al. (Oh *et al.*, 2010), they suggested a molecular mechanism throughout G protein-coupled receptor 120 (GPR120) functions as omega-3 fatty acids receptor/sensor. In other words, omega-3 fatty acids increase gene expression of GPR120 receptor and association with b-arrestin 2, as well as the association between Tak/Tab proteins in the adipose tissue of the treated subjects, improving markers of insulin metabolism (Oh *et al.*, 2010).

The current study demonstrated that compared with the placebo, omega-3 fatty acids supplementation for 12 weeks among subjects with DFU was associated with a significant decrease in serum hs-CRP, and significant increases in plasma TAC and GSH levels, but did not influence plasma NO and MDA concentrations. We have previously shown that 1000 mg

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of omega-3 fatty acids supplementation in gestational diabetes mellitus (GDM) women for 6 weeks had beneficial effects on maternal serum hs-CRP and plasma MDA concentrations, but did not affect other biomarkers of oxidative stress (Jamilian *et al.*, 2016). In addition, supplementation with omega-3 fatty acids in women with systemic lupus erythematosus for 12 weeks led to a significant decrease of CRP levels, but did not alter serum concentrations of IL-6 and IL-10 (Borges *et al.*, 2016). However, supplementation with omega-3 fatty acid did not change serum CRP concentrations among subjects with chronic periodontitis after 6-week and 12-week treatment (Deore *et al.*, 2014). Furthermore, no significant effect in TAC concentrations was seen following supplementation with a combined dietary supplementation containing omega-3, vitamin E, niacin and gamma-oryzanol among dyslipidemic patients (Accinni *et al.*, 2006). Different our findings with others might be explained by different study designs, the variation in the population studied, and the source of omega-3 fatty acids, dosage of omega-3 fatty acids as well as duration of the study. Omega-3 fatty acids intake directly decreases the production of inflammatory cytokines (Li *et al.*, 2005). Moreover, omega-3 fatty acids may decrease inflammation and oxidative stress through inhibiting activation of nuclear factor kappa-light-chain-enhancer of activated B cells (Rossi *et al.*, 2000; Wu *et al.*, 2004).

This trial had few limitations. Firstly, we did not evaluate gene expression of related to insulin, lipid and inflammation to explore the plausible mechanism. In addition, we did not evaluate other biomarkers of inflammation and oxidative stress. It must be kept in mind that in the current study, we quantified ulcer size using wound surface area measurements. The use of greatest length and width of the wound at each examination (Jessup, 2006) and the calculation of the ratio of greatest length and width (Gorin *et al.*, 1996) provide a coarse surface area change measure that has been used as an indicator of healing rate. We did not

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measure a wound area using more accurate determination of ulcer size, including a permanent marker, acetate or tracing paper (or a wound dressing packet), digital planimetry, digital photography combined with wound analysis software and a piece of graph paper. This is very unfortunate. Therefore, our findings should be interpreted with caution. Furthermore, in the current study, we quantified serum hs-CRP levels as a systemic inflammatory marker. Systemic markers such as CRP can be influenced by a variety of factors that are not always related to wounds. Wound specific measures of wound inflammatory markers such as interleukins or matrix metalloproteinases would have given more validity to the results. Therefore, this should be considered in our interpretation. In addition, presentation of wound healing trajectories in graphical form (time vs. % decrease in wound size for example) would have been useful.

Overall, taking omega-3 fatty acids for 12 weeks among subjects with DFU had beneficial effects on parameters of ulcer size, markers of insulin metabolism, serum hs-CRP, plasma TAC and GSH values; but, it did not had any effect on lipid profiles and other biomarkers of inflammation and oxidative stress. In addition, flaxseed oil omega-3 fatty acids may have played an indirect role in wound healing due to its effects on improved metabolic profiles.

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Authors' contributions

ZA contributed in conception, design, statistical analysis and drafting of the manuscript. ZS, FH, FB, MT and M-RM contributed in data collection and manuscript drafting. All authors approved the final version for submission.

Conflicts of interest

None.

Clinical trial registration number

http://www.irct.ir: IRCT201604025623N72.

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Table 1 General characteristics of study participants

	Placebo group (n=30)	Omega-3 group (n=30)	P[a]
Gender (%)			
Male	23 (76.7)	23 (76.7)	1.00†
Female	7 (23.3)	7 (23.3)	
Ulcer location			
Big toe	16 (53.3)	17 (56.7)	
Forefoot (Metatarsal 1-5)	11 (36.7)	10 (33.3)	0.99†
Heel	2 (6.7)	2 (6.7)	
Mid foot	1 (3.3)	1 (3.3)	
Ulcer duration (weeks)	3.4±0.8	3.3±0.9	0.56
CKD (%)	20 (66.7)	19 (63.3)	0.78†
Age (y)	59.9±9.2	58.8±11.2	0.66
Height (cm)	167.7±8.1	169.4±8.5	0.41
Weight at study baseline (kg)	75.8±9.1	77.3±11.8	0.56
Weight at end-of-trial (kg)	76.0±9.1	77.9±12.9	0.50
Weight change (kg)	0.2±1.8	0.6±3.2	0.58
BMI at study baseline (kg/m^2)	26.9±2.7	27.0±4.5	0.96
BMI at end-of-trial (kg/m^2)	27.0±2.8	27.2±4.6	0.88
BMI change (kg/m^2)	0.1±0.6	0.2±1.0	0.65
MET-h/day at study baseline	26.2±2.7	27.0±2.9	0.27
MET-h/day at end-of-trial	26.1±2.8	27.0±2.9	0.18
MET-h/day change	-0.1±0.7	0.04±0.9	0.36
Insulin therapy (%)	30 (100.0)	30 (100.0)	1.00†
Metformin therapy (%)	27 (90.0)	27 (90.0)	1.00†

Data are means± SDs.

[a] Obtained from independent t test. CKD, chronic kidney disease; METs, metabolic equivalents.

† Obtained from Pearson Chi-square test.

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Table 2 Dietary intakes of study participants throughout the study

	Placebo group (n=30)	Omega-3 group (n=30)	P[a]
Energy (kcal/d)	2344±139	2326±190	0.66
Carbohydrates (g/d)	320.3±33.6	318.2±46.7	0.84
Protein (g/d)	87.1±14.4	85.5±23.9	0.77
Fat (g/d)	82.8±14.5	82.6±19.5	0.96
SFAs (g/d)	24.6±6.4	24.7±7.6	0.97
PUFAs (g/d)	26.7±5.6	26.1±5.5	0.64
MUFAs (g/d)	21.9±5.5	22.8±8.0	0.62
Cholesterol (mg/d)	224.8±124.9	240.2±165.1	0.68
TDF (g/d)	17.8±3.2	18.5±4.7	0.53
Omega-3 (g/d)	1.0±0.4	1.0±0.3	0.45
Omega-6 (g/d)	23.5±3.6	24.1±4.2	0.52

Data are means± SDs.

[a] Obtained from independent t test.

MUFAs, monounsaturated fatty acids; PUFAs, polyunsaturated fatty acids; SFAs, saturated fatty acids; TDF, total dietary fiber.

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Table 3 Wound healing, metabolic profiles, and biomarkers of inflammation and oxidative stress at the baseline and after 12 weeks of the intervention in patients with diabetic foot ulcer

	Placebo group (n=30)			Omega-3 group (n=30)			
	Baseline	End-of-trial	Change	Baseline	End-of-trial	Change	P[a]
Ulcer length (cm)	3.4±1.7	2.4±1.8	-1.0±1.1	3.5±2.3	1.4±1.0	-2.1±2.3	0.03
Ulcer width (cm)	2.9±1.3	1.9±1.4	-1.0±1.0	2.9±2.1	1.1±1.0	-1.8±1.7	0.02
Ulcer depth (cm)	1.3±0.6	0.8±0.5	-0.5±0.5	1.4±0.5	0.5±0.4	-0.8±0.6	0.01
Ulcer symptoms							
Erythema (%)	30 (100.0)	14 (46.7)	-	30 (100.0)	7 (23.3)	-	0.05†
Discharge (%)	30 (100.0)	8 (26.7)	-	30 (100.0)	4 (13.3)	-	0.19†
Necrosis (%)	10 (100.0)	4 (40.0)	-	10 (100.0)	2 (20.0)	-	0.32†
FPG (mg/dL)	203.8±87.6	193.7±85.5	-10.1±45.8	199.5±99.7	160.9±80.8	-38.6±82.2	0.10
Insulin (μIU/mL)	16.6±9.3	17.9±8.5	1.4±8.3	15.8±9.2	11.4±6.7	-4.4±5.5	0.002
HOMA-IR	8.4±6.5	9.4±7.1	1.0±5.0	7.7±6.1	5.5±4.4	-2.1±3.0	0.005
QUICKI	0.29±0.02	0.28±0.02	-0.005±0.02	0.30±0.03	0.31±0.03	0.01±0.01	0.002
HbA1c (%)	7.9±0.7	7.8±0.7	-0.1±0.4	7.5±1.5	6.6±1.4	-0.9±1.5	0.01
Triglycerides (mg/dL)	136.5±41.8	139.4±46.4	2.9±18.3	146.1±73.1	128.4±56.1	-17.7±65.9	0.10
VLDL-cholesterol (mg/dL)	27.3±8.3	27.9±9.3	0.6±3.7	29.2±14.6	25.7±11.2	-3.5±13.1	0.10
Total cholesterol (mg/dL)	161.2±52.4	168.4±53.7	7.2±28.0	142.9±52.7	148.1±39.9	5.2±38.6	0.82

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LDL-cholesterol (mg/dL)	101.5±43.2	104.7±45.2	3.1±24.7	83.6±45.8	90.7±36.9	7.0±35.5	0.62
HDL-cholesterol (mg/dL)	32.4±6.9	35.8±7.9	3.4±6.5	30.0±4.4	31.7±5.7	1.7±3.6	0.20
hs-CRP (µg/mL)	47.8±32.1	39.6±26.1	-8.2±18.9	43.1±40.8	17.7±20.6	-25.5±31.5	0.01
NO (µmol/L)	42.7±3.9	45.0±6.8	2.3±7.7	46.3±4.7	46.1±5.5	-0.2±7.2	0.19
TAC (mmol/L)	957.1±219.1	883.7±292.2	-73.4±195.5	876.6±161.9	960.2±148.0	83.5±111.7	<0.001
GSH (µmol/L)	541.3±98.8	525.8±93.0	-15.5±129.7	526.0±100.1	586.7±114.9	60.7±140.2	0.03
MDA (µmol/L)	3.1±0.7	3.0±0.7	-0.1±0.4	2.6±0.4	2.5±0.3	-0.1±0.6	0.94

All values are means± SDs.

[a] P values represent the time × group interaction (computed by analysis of the one-way repeated measures ANOVA).

† Obtained from Chi-square test.

FPG, fasting plasma glucose; GSH, total glutathione; HOMA-IR, homeostasis model of assessment-estimated insulin resistance; hs-CRP, high-sensitivity C-reactive protein; MDA, malondialdehyde; NO, nitric oxide; QUICKI, quantitative insulin sensitivity check index; TAC, total antioxidant capacity.

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Table 4 Adjusted changes in metabolic profile of the patients with diabetic foot ulcer

	Placebo group (n=30)	Omega-3 group (n=30)	P[a]
Ulcer length (cm)	-1.1±0.2	-2.0±0.2	0.006
Ulcer width (cm)	-1.0±0.2	-1.8±0.2	0.004
Ulcer depth (cm)	-0.5±0.1	-0.8±0.1	0.008
FPG (mg/dL)	-8.6±10.5	-40.1±10.5	0.03
Insulin (μIU/mL)	1.4±1.0	-4.5±1.0	<0.001
HOMA-IR	1.0±0.7	-2.2±0.7	0.001
QUICKI	-0.006±0.004	0.01±0.004	<0.001
HbA1c (%)	-0.02±0.2	-1.0±0.2	<0.001
Triglycerides (mg/dL)	0.7±7.5	-15.5±7.5	0.13
VLDL-cholesterol (mg/dL)	0.1±1.5	-3.1±1.5	0.13
Total cholesterol (mg/dL)	9.7±5.5	2.7±5.5	0.38
LDL-cholesterol (mg/dL)	5.8±5.1	4.4±5.1	0.83
HDL-cholesterol (mg/dL)	3.7±1.0	1.5±1.0	0.11
hs-CRP (μg/mL)	-6.9±3.1	-26.8±3.1	<0.001
NO (μmol/L)	0.3±1.2	1.7±1.2	0.42
TAC (mmol/L)	-67.6±29.6	77.7±29.6	0.001
GSH (μmol/L)	-6.7±19.0	51.9±19.0	0.03
MDA (μmol/L)	0.1±0.1	-0.2±0.1	0.15

All values are means± SEs. Values are adjusted for baseline values, age and BMI at baseline.

[a] Obtained from ANCOVA.

FPG, fasting plasma glucose; GSH, total glutathione; HOMA-IR, homeostasis model of assessment-estimated insulin resistance; hs-CRP, high-sensitivity C-reactive protein; MDA, malondialdehyde; NO, nitric oxide; QUICKI, quantitative insulin sensitivity check index; TAC, total antioxidant capacity.

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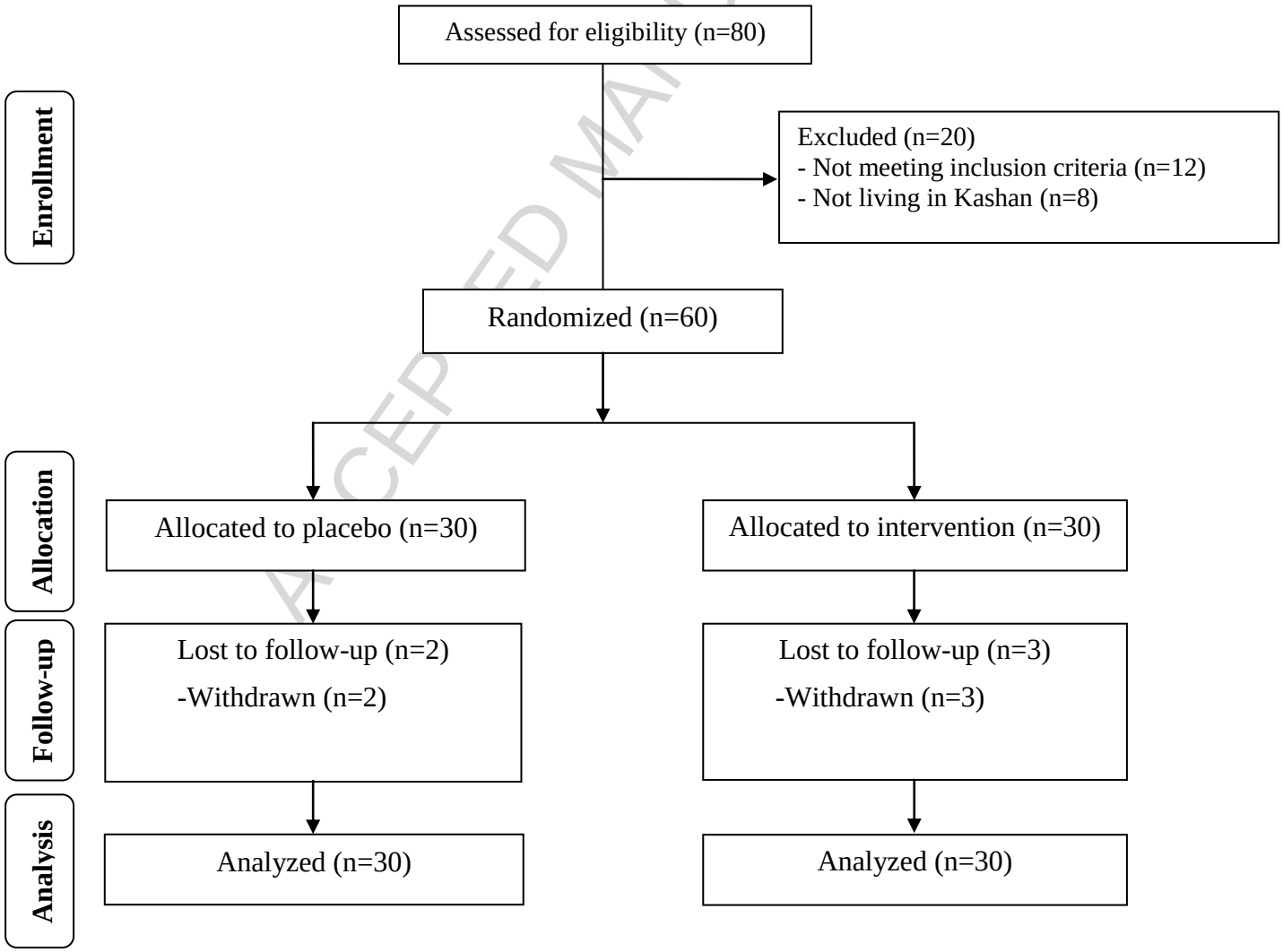
Legend to figure:

Fig. 1. Summary of patient flow diagram.



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Highlights

1. This study was conducted to determine the effects of omega-3 supplementation on wound healing and metabolic status in patients with grade 3 diabetic foot ulcer (DFU).

2. Omega-3 supplementation for 12 weeks in patients with DFU resulted in significant decreases in ulcer length, width and depth.

3. Omega-3 supplementation had beneficial effects on glucose metabolism in subjects with DFU.

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